■# PAPER_358 — AT2024tvd Wandering Massive Black Hole TDE: Off-Nuclear Disruption Physics

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Session: *97*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF off-nuclear wandering massive black hole TDE — frictional timescale and tidal radius **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

AT2024tvd is the most compelling observed wandering massive black hole (wMBH) caught in the act of tidally disrupting a star at projected physical offset $r_{offset} = 2.47\times10^{1}$■ *m from the host galaxy nucleus. UQFF computes the tidal radius* $r_{tide} = R_{star}\cdot(M_{BH}/M_{star})$^(1/3), *the dynamical friction timescale* t_{fric} for the wMBH sinking to the nucleus, and the full F*U*Bi_i at the off-nuclear disruption site.

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2. Core Physics

2.1 Off-Nuclear Disruption Offset

```
r_{\rm offset} = 2.47 \times 10^{17}\ \mathrm{m} \approx 8.0\ \mathrm{pc}
```

This is the projected distance between AT2024tvd and the host nucleus, constraining the wandering distance of the massive black hole.

2.2 Tidal Disruption Radius

```
r_{\rm tide} = R_\star \left(\frac{M_{\rm BH}}{M_\star}\right)^{1/3}
```

For a solar-like star disrupted by a black hole of mass M_BH ~ 10■ M■:

```
r_{\rm tide} = 7 \times 10^8 \times \left(\frac{10^6 M_\odot}{M_\odot}\right)^{1/3}\
\mathrm{m} = 7 \times 10^8 \times 100 = 7 \times 10^{10}\ \mathrm{m} \approx 0.5\ R_\odot
```

2.3 Dynamical Friction Timescale

```
t_{\rm fric} = \frac{0.428}{\ln\Lambda} \cdot \frac{M_{\rm host}}{M_{\rm BH}} \cdot
\frac{r_{\rm offset}^2}{\sigma_\star^2} \cdot \frac{1}{r_{\rm offset}}
```

Simplified:

```
t_{\rm fric} = \frac{0.428}{\ln\Lambda} \cdot \frac{r_{\rm offset}}{v_c} \cdot
\frac{M_{\rm host}}{M_{\rm BH}}
```

For $r_{offset} = 8$ *pc*, $v_c \sim 200$ km/s, $M_{host}/M_{BH} \sim 10^3$:

```
t_{\rm fric} \sim 10^8 - 10^9\ \mathrm{yr}
```

2.4 UQFF FUBi_i at Off-Nuclear Site

```
F_{U\_Bi\_i}(r_{\rm offset}) = F_{U\_Bi\_i}(M_{\rm BH}) \cdot \left(\frac{r_{\rm
tide}}{r_{\rm offset}}\right)^2
```

The r^2 dependence reflects force dilution with the much larger offset distance.

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3. Key Values

Quantity	Formula	Value
r_offset	JWST observation	2.47×10^{1}■ m ≈ 8 pc
r_tide	R∗·(*M*BH/M_∗)^(1/3)	~0.5 AU
t_fric	Chandrasekhar formula	10■–10■ yr
M_BH	Spectral fit	~10■ M■

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4. Physical Significance

AT2024tvd is the first confirmed off-nuclear TDE with a massive BH at > 1 pc offset. The UQFF framework predicts that the vacuum buoyancy field F*U*Bi*i is local — it is set by M*BH and r*tide, not by the nuclear distance. This means off-nuclear wMBH TDEs have the same F*U*Bi*i value as nuclear TDEs of the same BH mass, a testable prediction: UQFF force amplitude should correlate with M*BH (not with r*offset).

The t_fric ~ 10■–10■ yr frictional timescale implies wMBHs are common during the galaxy assembly epoch, and UQFF predicts their spatial distribution modifies the ICM density on 10–100 pc scales.

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5. Deduplication Note

vs. PAPER_351 (ASASSN-14li): Both are TDEs; ASASSN-14li is nuclear. AT2024tvd is off-nuclear, introducing r_offset and t_fric which are absent in PAPER_351.

Unique: First off-nuclear wMBH TDE in the UQFF dataset.

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6. Classification

Physics Territory: FIRST UQFF off-nuclear wandering MBH TDE — r*tide* + *t*fric + F*U*Bi_i(offset) **Scale:** Sub-galactic (8 pc offset from nucleus) **CP Implementation:** `AT2024tvdWanderingMBHTDECalculator` (CondensedPhysics4.py, Session 97)
